Bolton Howes

Assistant Professor & Director of the Wheaton College Field Station (Black Hills, SD)
Wheaton College

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Professional Experience

2025–	**Wheaton College** Assistant Professor & Director of the Wheaton College Field Station (Black Hills, SD)
2024–2025	**University of Washington & Western Washington University** NSF Postdoctoral Fellow Advisors: Akshay Mehra and Brady Foreman
2023–2024	**University of Miami (Rosenstiel School of Marine, Atmospheric, and Earth Science)** Postdoctoral Researcher, Marine Geology Advisor: Sam Purkis

Education

2017–2023	**Princeton University**, Princeton, NJ Ph.D., Geosciences Dissertation: Developing physical paleoenvironmental proxies: from ooids to extinction Advisor: Adam Maloof
2015–2017	**University of Georgia**, Athens, GA M.Sc., Geology Thesis: Sequence stratigraphic expression of flexural subsidence: Middle Jurassic Twin Creek Limestone, Wyoming, U.S.A. Advisor: Steven Holland
2011–2015	**Macalester College**, St. Paul, MN B.A., Geology Advisor: Raymond Rogers

Awards and Honors

2024	UW eScience Postdoctoral Fellowship [$4,000]
2023	NSF EAR Postdoctoral Fellowship [$180,000]
2017–2022	Department of Geosciences Graduate Research Fund [3x$5,000]
2017	Outstanding M.S. Student of the Year, Department of Geology, University of Georgia
2015–2017	Graduate School Assistantship, University of Georgia (university-wide competition; department nomination, faculty-panel selection)
2016	Graduate Research Grant, Geological Society of America
2016	Miriam Watts-Wheeler Award, University of Georgia [$1,600]
2015	Henry Lepp Award for Dedication to Scientific Research, Macalester College

Publications (* denotes student advisee)

In review

12. Filip Novak*, **Bolton Howes**, Henry Yuan, Brady Foreman, Akshay Mehra. From outcrop to point cloud: A three-dimensional technique for the remote analysis of terrestrial and extraterrestrial sedimentary structures. In review at *GSA Today*.

11. Anthony Semeraro, Benjamin Cardenas, **Bolton Howes**, Brady Foreman. Reconstructing river planform morphology in alluvial stratigraphy using paleocurrents. In review at *The Depositional Record*.

Published

10. Morgan Chakraborty, Hannah Shernisky, Clément Pollier, Amanda Oehlert, Arash Sharifi, Ali Pourmand, **Bolton Howes**, Benjamin Minch, Mohammad Moniruzzaman, Larissa Frühe, Mattie Rodrigue, Sam Purkis. Brine pool microbes enrich metalliferous sediments in salt giant basins. 2026. *AGU Advances*. doi:10.1029/2026AV002570

9. Kevin Vélez-Rosado, Olga Zalles-Grebetskaya, Jeffrey Wilson Mantilla, Blair Schoene, Adam Maloof, **Bolton Howes**. New material of *Dolichochampsa minima* (Archosauria: Crocodylia) from the Cretaceous–Palaeogene El Molino Formation of Bolivia sheds light on the early evolution of Gavialinae. 2025. *Journal of Systematic Palaeontology*. doi:10.1080/14772019.2025.2496524

8. Morgan Chakraborty, Arash Sharifi, Francesca Benzoni, François Tissot, Ali Pourmand, Marco Taviani, **Bolton Howes**, Peter Swart, Chaojin Lu, Mattie Rodrigue, Sam Purkis. Deep-water corals indicate the Red Sea survived the last glacial lowstand. 2025. *PNAS*. doi:10.1073/pnas.2415559122

7. Sam Purkis, Steven Ward, **Bolton Howes**, Jake Longenecker, Morgan Chakraborty, Akos Kalman, Amy Clement, Arash Sharifi, Francesca Benzoni, Christopher Clarke, Mattie Rodrigue. A 1600-year record of extreme rainfall in northern Arabia. 2025. *Science Advances*. doi:10.1126/sciadv.adq3173

6. **Bolton Howes**, Akshay Mehra, Emily Geyman, Julia Wilcots, Ryan Manzuk, Curtis Deutsch, Adam Maloof. The where, when, and how of ooid formation: what ooids tell us about ancient seawater chemistry. 2024. *Earth and Planetary Science Letters*. doi:10.1016/j.epsl.2024.118697

5. Gaëlle Duchâtellier, Amanda Oehlert, Hannah Shernisky, Clément Pollier, Peter Swart, **Bolton Howes**, Sam Purkis. Sedimentary porewaters record regional tectonic and climate events that perturbed a deep-sea brine pool in the Gulf of Aqaba, Red Sea. 2024. *Science of the Total Environment*. doi:10.1016/j.scitotenv.2023.168804

4. Akshay Mehra, **Bolton Howes**, Ryan Manzuk, Alex Spatzier, Bradley Samuels, Adam Maloof. A novel technique for producing three-dimensional data using serial sectioning and semi-automatic image classification. 2022. *Microscopy and Microanalysis*. doi:10.1017/S1431927622012442

3. Emily Geyman, Ziman Wu, Matthew Nadeau, Stacey Edmonsond, Andrew Turner, Sam Purkis, **Bolton Howes**, Blake Dyer, Anne-Sofie Ahm, Nan Yao, Curtis Deutsch, John Higgins, Daniel Stolper, Adam Maloof. The origin of carbonate mud and implications for global climate. 2022. *PNAS*. doi:10.1073/pnas.2210617119

2. **Bolton Howes**, Akshay Mehra, Adam Maloof. Three-dimensional morphometry of ooids in oolites: a new tool for more accurate and precise paleoenvironmental interpretation. 2021. *Journal of Geophysical Research: Earth Surface*. doi:10.1029/2020JF005601

1. Paul Hoffman, Galen Halverson, Daniel Schrag, John Higgins, . . . , **Bolton Howes**, et al. Snowballs in Africa: sectioning a long-lived Neoproterozoic carbonate platform and its bathyal foreslope (NW Namibia). 2021. *Earth-Science Reviews*. doi:10.1016/j.earscirev.2021.103616

In prep.

Filip Novak*, **Bolton Howes**, Brady Foreman, Akshay Mehra. Minimal hydrologic changes spanning the Paleocene-Eocene Thermal Maximum: Biased proxy or buffered signal?

Bolton Howes, Morgan Chakraborty, Sam Purkis. The afterlife of lowstand sediments.

Bolton Howes, Delaney Todd, Katherine Sullivan, Akshay Mehra, Brady Foreman. Increased river mobility and floodplain erosion during the Paleocene-Eocene global warming event.

Conference Abstracts (* denotes student advisee)

2026	Filip Novak*, **Bolton Howes**, Brady Foreman, Akshay Mehra. Cross-sets as a Paleohydrologic Proxy: Insights from the Paleocene-Eocene Thermal Maximum of the Bighorn Basin, Wyoming. *GSA* (2026).

2025	Filip Novak*, **Bolton Howes**, Brady Foreman, Akshay Mehra. From Outcrop to Point Cloud: Using Three-Dimensional Techniques to Study Fluvial Dynamics During the Paleocene-Eocene in Wyoming. *GSA* (2025).
2025	**Bolton Howes**, Anthony Semeraro, Akshay Mehra, Brady Foreman. Reconstructing river planform morphology. *Geobiology* (2025).
2022	**Bolton Howes**, Ryan Manzuk, Akshay Mehra, Adam Maloof. Tales of transport and chemistry from three-dimensional growth histories of ooids. *AGU Fall Meeting* (Chicago, Illinois).
2022	Ryan Manzuk, **Bolton Howes**, Emily Geyman, Indu Panigrahi, Dave Singh, Adam Maloof. Multispectral petrography image analysis as a method for paleoenvironmental and paleoecological reconstruction; applications to Earth's first reefs. *AGU Fall Meeting* (Chicago, Illinois).
2022	Cedric Hagen, John Higgins, Adam Maloof, Stacey Edmonsond, Ryan Manzuk, **Bolton Howes**, Pingping Zhao. Machine learning and computational image analysis leveraged to probe relationships between carbonate isotopic composition, diagenetic alteration, and sedimentary facies isotopic variability. *AGU Fall Meeting* (Chicago, Illinois).
2022	Adam Maloof, Ryan Manzuk, Emily Geyman, Akshay Mehra, Jaap Kaandorp, Mark Webster, Stacey Edmonsond, **Bolton Howes**, and Cedric Hagen. From modern analogs to three dimensions: Lessons learned for interpreting the stratigraphic record of the Proterozoic–Phanerozoic transition. *GSA Annual Meeting* (Denver, Colorado).
2018	**Bolton Howes**, Akshay Mehra, and Adam Maloof. Three-dimensional reconstructions of Holocene and Neoproterozoic oolites to measure porosity, permeability, and volume-shape evolution of ooids. *GSA Annual Meeting* (Indianapolis, Indiana).
2014	**Bolton Howes** and Raymond Rogers. Revisiting a regionally significant terrestrial bounding surface in the Upper Cretaceous (Campanian) Two Medicine Formation, northwestern Montana. *GSA Annual Meeting* (Vancouver, British Columbia).

Invited Talks

2026	University of Chicago
2026	University of Washington College of the Environment, Pathways to a Professor Panel
2025	Purdue University
2025	University of Washington, PaleoLunch
2025	Macalester College, Sed/Strat Seminar
2025	Western Washington University, Department Seminar
2024	Wheaton College, Department Seminar
2020	Macalester College, Graduate School Open House
2018	Macalester College, Sed/Strat Seminar

Fieldwork Experience (84 Weeks)

2026	Bolivia	2 weeks	K–Pg paleontology
2024	Indonesia	2 weeks	Mapping seafloor ecology and natural hazards
2024 & 2025	NW Wyoming	5 weeks	The fluvial response to the PETM
2023	Red Sea	3 weeks	Geomorphology and stratigraphy to evaluate natural hazard risk
2019 & 2022	Bolivia	12 weeks	The K–Pg Boundary in lacustrine and fluvial settings
2021	southern Utah	2 weeks	Cyclicity in the Late Paleozoic Ice Age
2019	Calabria, Italy	1 week	Measuring climate change in olive orchards
2019	Yukon, Canada	5 weeks	Archaeocyathid reefs and the Cambrian Explosion
2018	Utah	1 week	Measuring climate change (GEOL 201 course fieldwork)
2018	Spain and Italy	6 weeks	The K–Pg Boundary and slope sediments

2018	southern Namibia	4 weeks	Neoproterozoic and Paleozoic glacial deposits
2018	Western Australia	1 week	Modern and recent carbonate platforms
2018	Nevada/Arizona	3 weeks	Stratigraphy and geochemistry of the Late Paleozoic Ice Age
2017	northern Namibia	6 weeks	Field assistant to Paul Hoffman. Mapping Snowball Earth
2016	western Wyoming	6 weeks	Sequence stratigraphy of the Twin Creek Limestone
2014	NW Montana	2 weeks	Sequence stratigraphy and stratigraphic paleobiology, Two Medicine Fm.
2014	Patagonia, Chile	12 weeks	Restoration ecology and conservation biology in the Aysén Region
2013 & 2014	NW Wyoming	5 weeks	Terrestrial stratigraphy and paleobiology of the Cloverly Formation
2014	Tall Dhiban, Jordan	6 weeks	Archeological excavation: Early Bronze Age through the 19th century

Graduate Committees

2026	Katherine Izzo	M.Sc. Committee Member, Western Washington University

Undergraduates Advised

2026	Carson Frear	Bolivian K–Pg
2024–2025	Filip Novak	Photogrammetry and fluvial sedimentology
2021	Anubhav Agarwal	Transfer learning for geological image segmentation
2021	Hugh Shields	Transfer learning for geological image segmentation
2021	Anna Krokhine	Numerical models of ooid growth
2020	Emilio Cano-Renteria	Geological image segmentation and ooid growth
2019	Sarah Brown	K–Pg boundary stratigraphy of Bolivia
2018	Celia Aranda-Reina	Climate variability during the K–Pg in Spain

Undergraduates Involved in Fieldwork

2026	Carson Frear	2 weeks	Bolivia
2025	Filip Novak	2 weeks	Wyoming
2019	10 Undergraduates	1 week	Italy
2019	Sarah Brown	5 weeks	Bolivia
2019	Galen Cadley	5 weeks	Yukon, Canada
2018	8 Undergraduates	1 week	Utah
2018	Celia Aranda-Reina	4 weeks	Spain
2016	Tony Moraes	4 weeks	western Wyoming

Teaching

Instructor of Record

Spr. 2027	**GEOL 212:** The Dynamic Earth and Environment, Wheaton College
Spr. 2026	**GEOL 321:** Earth History and Stratigraphy, Wheaton College

Laboratory Instructor

Spr. 2026	**GEOL 212L:** The Dynamic Earth and Environment (lab section), Wheaton College
Fall 2025	**GEOL 212L:** The Dynamic Earth and Environment (lab section), Wheaton College

Teaching Assistant

Spr. 2021	**ENV 354-GEO 368** Climate and Weather: Order in the Chaos, Princeton University
Fall 2019*	**FRS 161:** Earth: Crops, Culture, and Climate (in Italy), Princeton University
Fall 2018*	**GEOL 201:** Measuring Climate Change: Data Analysis and Science Writing, Princeton University
2016–2017	Geology Tutor, University of Georgia Student-Athlete Academic Center
Spr. 2015	**GEOL 165:** History and Evolution of the Earth, Macalester College
Fall 2014	**GEOL 300:** Paleobiology, Macalester College

* *Indicates courses that included a fieldwork component consisting of a week-long trip (Utah and Italy) to make observations for research projects. My teaching duties included preparing supplies and logistics, and helping students with both data collection—especially with drones, differential GPS, and surveying equipment—and data analysis. The data collected during fieldwork was part of a semester-long project.*

Leadership and Service

2023	Fellow, Scholars Institute Fellows Program (mentoring first-generation students), Princeton University
2018–2021	Residential Graduate Student, Princeton University
2016–2017	Student Volunteer, Geological Society of America
2015	Leadership Committee of Macalester College Fellowship of Christian Athletes
2012–2013	Resident Assistant, Macalester College
2011–2014	Captain of Macalester College Football Team

Professional Activities

Convener	2019 GSA Annual Meeting technical session: Hello (ancient) world!: Exploring the Neoproterozoic to Cambrian interval by quantitatively probing the rock record
Reviewer	NSF EAR Geoinformatics, Geophysical Research Letters

Public Engagement

2025	*Tiny Living Beings* podcast (American Society for Microbiology): "4.5 Billion Years of Earth History in One Hour"